

Solutions to Problem Set 1

1. Elementary combinatorics

Use the results on elementary combinatorics seen in the lecture to answer the following questions. Explain your reasoning.

- (a) How many possibilities are there to distribute 12 indistinguishable coins to 5 different persons?
- (b) Assume you have 9 colors to mark 5 different courses you have taken this term in your notes. How many possibilities to choose these colors are there, if every subject has its distinct color.
- (c) There are 10 lamps in a room, that can be turned on or off independently from each other. How many possibilities are there for the configuration of lights?
- (d) In a lottery, 6 out of 49 numbers are chosen at once. How many possibilities are there to guess exactly 3 out of these numbers correctly? Hence, calculate the probability to guess 3 out of 6 numbers correctly in this type of lottery.
- (e) There are 12 female and 8 male candidates for a committee made out of 8 persons in total. How many possibilities are there for the committee if there have to be at least 3 female and at most 5 male committee members?
- (f) On the unit circle, $n \geq 2$ equidistant points are put. Out of those, we choose two of them uniformly at random. Calculate the probability p_n that the chosen points are neighbors.
- (g) How many possible anagrams can be formed out of the name "HANNAH"? How many of those are palindromes (i.e. coinciding with the respective word read backwards)?
- (h) How many possibilities are there to order $n \geq 1$ books into a shelf, if $1 \leq m \leq n$ of the books are indistinguishable?

- (a) This corresponds to choosing $r = 12$ elements (the coins) out of a set with $n = 5$ elements (the persons) *unordered and with repetition*. Therefore we have

$$\binom{n+r-1}{r} = \binom{5+12-1}{12} = \binom{16}{12} = 1820$$

possibilities for the distribution.

- (b) In this case, we are choosing $r = 5$ elements (the courses) out of a set with $n = 9$ elements (the colors) *ordered but without repetition*. Thus, there are

$$\frac{n!}{(n-r)!} = \frac{9!}{(9-5)!} = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 15120$$

possibilities.

- (c) Every lamp can be turned on or off, so we are dealing with choosing $r = 10$ (the lamps) elements out of a set with $n = 2$ elements (the configuration on / off) *ordered and with repetition*. In other words, there are

$$n^r = 2^{10} = 1024$$

possibilities.

- (d) The 49 numbers can be decomposed into 43 losing and 6 winning numbers. Thus the problem is broken down to choosing 3 out of 43 numbers *unordered and without repetition* and 3 out of 6 numbers *unordered and without repetition*. In other words, there are

$$\binom{43}{3} \cdot \binom{6}{3} = 2468290.$$

Note that there are $\binom{49}{6}$ ways in total to draw 6 out of 49 numbers *unordered and without repetition*. The probability to obtain exactly 3 correct guesses (assuming a fair draw, where every outcome is equally likely) is therefore

$$\mathbf{P}[\text{"3 correct guesses"}] = \frac{\binom{43}{3} \cdot \binom{6}{3}}{\binom{49}{6}} \approx 0.0176 = 1.76\%.$$

- (e) The committee can consist of j female and $8 - j$ male members, where $j \in \{3, \dots, 8\}$. Similarly as in the previous problem, choosing j female out of 12 candidates and $8 - j$ out of 8 male candidates corresponds to choosing *unordered and without repetitions*, giving $\binom{12}{j} \cdot \binom{8}{8-j}$ choices, so the total number is

$$\begin{aligned} \sum_{j=3}^8 \binom{12}{j} \cdot \binom{8}{8-j} &= \binom{12}{3} \cdot \binom{8}{5} + \binom{12}{4} \cdot \binom{8}{4} + \binom{12}{5} \cdot \binom{8}{3} + \binom{12}{6} \cdot \binom{8}{2} \\ &\quad + \binom{12}{7} \cdot \binom{8}{1} + \binom{12}{8} \cdot \binom{8}{0} \\ &= 124025. \end{aligned}$$

- (f) In the case $n = 2$, we trivially have $p_2 = 1$. From now on suppose that $n \geq 3$. The number of ways to choose 2 points out of n points on the circle *unordered and without repetition* is $\binom{n}{2}$. Out of those, the number of ways to choose points that are neighbors is simply n (indeed, to avoid overcounting, we consider groups $\{1, 2\}$, $\{2, 3\}$, ..., $\{n, 1\}$, i.e. looking at right neighbors). Therefore,

$$p_n = \frac{n}{\binom{n}{2}} = \frac{2n(n-2)!}{n!} = \frac{2}{n-1}, \quad n \geq 3.$$

- (g) The number of ways to rearrange the name "HANNAH" is

$$\frac{6!}{2! \cdot 2! \cdot 2!} = 90.$$

A palindrome must have the form $X_1 X_2 X_3 | X_3 X_2 X_1$ with $X_1, X_2, X_3 \in \{A, H, N\}$ all distinct, so there are

$$3! = 6$$

many palindromes that can be formed.

- (h) This corresponds to the ways of reordering the word

$$\underbrace{AA \dots A}_{m \text{ members}} B_1 B_2 \dots B_{n-m},$$

where A stands for the indistinguishable books. Thus, there are

$$\frac{n!}{m!}$$

many ways.

2. (*) Set operations and events, de Morgan's rules

[4 Points]

A (six-sided) die is rolled $N \in \mathbb{N}$ times. Let A_k be the event that the k -th role is a 3 and B_k is the event that the k -th roll is a 6 ($1 \leq k \leq N$).

- (a) Express with the help of the events A_k, B_k , as well as elementary set operations the following events:
- A : 6 is never obtained.
 - B : a 3 is obtained at least once.
 - C : a 6 is obtained at least once and a 3 is obtained at least twice.

(b) Describe the following events in words:

$$\left(\bigcup_{i=1}^N A_i^c\right)^c, \quad \bigcup_{i=1}^{N-2} (A_i \cap A_{i+1} \cap B_{i+2}), \quad \bigcap_{i=1}^{N-1} (A_i \cup B_{i+1}).$$

Now let $(A_i)_{i \in I}$ be a family of subsets of a non-empty set Ω . Prove *de Morgan's rules*:

(c) $\left(\bigcup_{i \in I} A_i\right)^c = \bigcap_{i \in I} A_i^c.$

(d) $\left(\bigcap_{i \in I} A_i\right)^c = \bigcup_{i \in I} A_i^c.$

(a) We have (for instance)

i. $A = \bigcap_{k=1}^N B_k^c,$

ii. $B = \bigcup_{k=1}^N A_k,$

iii. $C = \bigcup_{k=1}^N B_k \cap \bigcup_{\ell, m=1, \ell \neq m}^N (A_\ell \cap A_m).$

(b) The events correspond to:

- N times rolling a 3 (by de Morgan's rules),
- The sequence "3-3-6" appears at least once, and
- Every pair of consecutive numbers has either a 3 at the left or a 6 at the right.

Turning to de Morgan's rules, we have

(c)

$$\begin{aligned} \omega \in \left(\bigcup_{i \in I} A_i\right)^c &\Leftrightarrow \omega \notin \bigcup_{i \in I} A_i &\Leftrightarrow \omega \notin A_i \text{ for every } i \in I \\ &\Leftrightarrow \omega \in A_i^c \text{ for every } i \in I &\Leftrightarrow \omega \in \bigcap_{i \in I} A_i^c. \end{aligned}$$

(d)

$$\begin{aligned} \omega \in \left(\bigcap_{i \in I} A_i\right)^c &\Leftrightarrow \omega \notin \bigcap_{i \in I} A_i &\Leftrightarrow \omega \notin A_i \text{ for at least one } i \in I \\ &\Leftrightarrow \omega \in A_i^c \text{ for at least one } i \in I &\Leftrightarrow \omega \in \bigcup_{i \in I} A_i^c. \end{aligned}$$

3. (*) Rolling dice

[4 Points]

A fair die is rolled twice. Calculate the probability for the following events.

- The number 6 occurs exactly once.
- Both numbers are odd.
- The sum of the numbers is 4.
- The sum of the numbers is an integer multiple of 3.

Write down the probability space $(\Omega, \mathcal{F}, \mathbf{P})$ and the corresponding events explicitly.

The probability space we are using is

$$(\Omega, \mathcal{F}, \mathbf{P}) = (\{1, 2, 3, 4, 5, 6\}^2, \mathcal{P}(\{1, 2, 3, 4, 5, 6\}^2), \mathbf{P}), \quad \mathbf{P}[A] = \frac{|A|}{|\Omega|} = \frac{|A|}{36}.$$

(a) The event is given by

$$A = \{(1, 6), (2, 6), (3, 6), (4, 6), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5)\}.$$

$$\text{Therefore, } \mathbf{P}[A] = \frac{10}{36} = \frac{5}{18}.$$

(b) The event is given by

$$B = \{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5)\} \\ (= \{1, 3, 5\}^2).$$

$$\text{Therefore, } \mathbf{P}[B] = \frac{9}{36} = \frac{1}{4}.$$

(c) The event is given by

$$C = \{(1, 3), (2, 2), (3, 1)\}.$$

$$\text{Therefore, } \mathbf{P}[C] = \frac{3}{36} = \frac{1}{12}.$$

(d) The event is given by

$$D = \{(1, 2), (2, 1), (1, 5), (2, 4), (3, 3), (4, 2), (5, 1), (3, 6), (5, 4), (4, 5), (6, 3), (6, 6)\}.$$

$$\text{Therefore, } \mathbf{P}[D] = \frac{12}{36} = \frac{1}{3}.$$

4. Inclusion-Exclusion formula

Consider a probability space $(\Omega, \mathcal{F}, \mathbf{P})$. Let $(A_j)_{j=1}^n$ be events, $A_j \in \mathcal{F}$.

(a) Using induction prove:

$$\mathbf{P}\left[\bigcup_{j=1}^n A_j\right] = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbf{P}\left[\bigcap_{j=1}^k A_{i_j}\right].$$

(b) Using induction prove the following statements:

$$\mathbf{P}\left[\bigcup_{j=1}^n A_j\right] \leq \sum_{j=1}^n \mathbf{P}[A_j] - \sum_{j=1}^{n-1} \mathbf{P}[A_j \cap A_{j+1}], \\ \mathbf{P}\left[\bigcup_{j=1}^n A_j\right] \geq \sum_{j=1}^n \mathbf{P}[A_j] - \sum_{i,j=1, i < j}^n \mathbf{P}[A_j \cap A_i].$$

(a) We use induction. The base case is true, i.e., when $n = 1$, then

$$\mathbf{P}\left[\bigcup_{j=1}^1 A_j\right] = \mathbf{P}[A_1] = \sum_{1 \leq i_1 \leq 1} \mathbf{P}[A_{i_1}].$$

We consider now the inductive step. We have

$$\mathbf{P}\left[\bigcup_{j=1}^{n+1} A_j\right] = \mathbf{P}[A_{n+1}] + \mathbf{P}\left[\bigcup_{j=1}^n A_j\right] - \mathbf{P}\left[\left(\bigcup_{j=1}^n A_j\right) \cap A_{n+1}\right] \\ = \mathbf{P}[A_{n+1}] + \mathbf{P}\left[\bigcup_{j=1}^n A_j\right] - \mathbf{P}\left[\bigcup_{j=1}^n (A_j \cap A_{n+1})\right].$$

We then apply the induction hypothesis (for n) to both terms on the right-hand side, which yields

$$\mathbf{P}\left[\bigcup_{j=1}^{n+1} A_j\right] = \mathbf{P}[A_{n+1}] + \sum_{k=1}^n \left((-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_k}] \right) \\ - \sum_{k=1}^n \left((-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_k} \cap A_{n+1}] \right).$$

Rewriting the last summand gives

$$\begin{aligned} \mathbf{P} \left[\bigcup_{j=1}^{n+1} A_j \right] &= \mathbf{P}[A_{n+1}] + \sum_{k=1}^n \left((-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_k}] \right) \\ &\quad + \sum_{k=2}^{n+1} \left((-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_{k-1} \leq n} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_{k-1}} \cap A_{n+1}] \right). \end{aligned}$$

With this, we see that the last summand is exactly $(-1)^n \mathbf{P}[A_1 \cap \dots \cap A_{n+1}]$. We split this term off and find

$$\begin{aligned} \mathbf{P} \left[\bigcup_{j=1}^{n+1} A_j \right] &= \mathbf{P}[A_{n+1}] + (-1)^n \mathbf{P}[A_1 \cap \dots \cap A_{n+1}] \\ &\quad + \sum_{k=1}^n \left((-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_k}] \right) \\ &\quad + \sum_{k=2}^{n+1} \left((-1)^{k-1} \sum_{1 \leq i_1 < \dots < i_{k-1} \leq n} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_{k-1}} \cap A_{n+1}] \right) \\ &= \mathbf{P}[A_{n+1}] + (-1)^n \mathbf{P}[A_1 \cap \dots \cap A_{n+1}] \\ &\quad + \sum_{k=1}^n \left((-1)^{k-1} \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n+1 \\ n+1 \notin \{i_1, \dots, i_k\}}} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_k}] \right) \\ &\quad + \sum_{k=2}^{n+1} \left((-1)^{k-1} \sum_{\substack{1 \leq i_1 < \dots < i_k \leq n+1 \\ n+1 \in \{i_1, \dots, i_k\}}} \mathbf{P}[A_{i_1} \cap \dots \cap A_{i_k}] \right). \end{aligned}$$

We are done, since $\mathbf{P}[A_{n+1}]$ exactly corresponds to the “missing” summand in the second sum.

- (b) We want to use induction. The base case are clearly true, i.e., when $n = 1$ both formulas are true. For the inductive step suppose the formulas are true for an integer n and we will try to prove them for $n + 1$. The first one:

$$\begin{aligned} \mathbf{P} \left[\bigcup_{j=1}^{n+1} A_j \right] &= \mathbf{P} \left[\bigcup_{j=1}^n A_j \cup A_{n+1} \right] \\ &= \mathbf{P} \left[\bigcup_{j=1}^n A_j \right] + \mathbf{P}[A_{n+1}] - \mathbf{P} \left[\bigcup_{j=1}^n A_j \cap A_{n+1} \right] \\ &\leq \mathbf{P} \left[\bigcup_{j=1}^n A_j \right] + \mathbf{P}[A_{n+1}] - \mathbf{P}[A_n \cap A_{n+1}] \\ &\leq \sum_{j=1}^n \mathbf{P}[A_j] - \sum_{j=1}^{n-1} \mathbf{P}[A_j \cap A_{j+1}] + \mathbf{P}[A_{n+1}] - \mathbf{P}[A_n \cap A_{n+1}] \\ &= \sum_{j=1}^{n+1} \mathbf{P}[A_j] - \sum_{j=1}^n \mathbf{P}[A_j \cap A_{j+1}]. \end{aligned}$$

For the second formula we have:

$$\begin{aligned}
\mathbf{P} \left[\bigcup_{j=1}^{n+1} A_j \right] &= \mathbf{P} \left[\bigcup_{j=1}^n A_j \cup A_{n+1} \right] \\
&= \mathbf{P} \left[\bigcup_{j=1}^n A_j \right] + \mathbf{P}[A_{n+1}] - \mathbf{P} \left[\bigcup_{j=1}^n (A_j \cap A_{n+1}) \right] \\
&\geq \mathbf{P} \left[\bigcup_{j=1}^n A_j \right] + \mathbf{P}[A_{n+1}] - \sum_{j=1}^n P(A_j \cap A_{n+1}) \\
&\geq \sum_{j=1}^n \mathbf{P}[A_j] - \sum_{i,j=1, i < j}^n \mathbf{P}[A_j \cap A_i] + \mathbf{P}[A_{n+1}] - \sum_{j=1}^n P(A_j \cap A_{n+1}) \\
&= \sum_{j=1}^{n+1} \mathbf{P}[A_j] - \sum_{i,j=1, i < j}^{n+1} \mathbf{P}[A_j \cap A_i].
\end{aligned}$$